

## CHENZHOU, China

WMO: 579720

Lat: 25.7355N

Lon: 112.9773E

Elev: 1210

StdP: 14.06

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 30.7       | 32.6 | 22.1                          | 17.3 | 43.8 | 25.8 | 20.7 | 40.8 | 15.6                  | 46.8 | 12.5 | 47.4 | 2.8                   | 0    | 0.343 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.3                   | 96.8            | 76.9 | 94.8 | 76.7 | 92.9 | 76.5 | 79.7                | 89.6 | 79.0 | 88.7 | 78.2 | 87.8 | 5.4                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.4                            | 149.2 | 83.6 | 76.5 | 144.8 | 83.0 | 75.9 | 141.6 | 82.7 | 44.4          | 89.7 | 43.6 | 88.8 | 42.9 | 88.0 | 86.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 19.5              | 14.4 | 11.1 | 28.9                       | 99.2 | 2.1                | 2.3 | 27.3                                               | 100.9 | 26.1       | 102.2 | 24.9       | 103.5 | 23.3       | 105.1 |
|                   |      |      | 27.2                       | 81.5 | 2.4                | 2.5 | 25.4                                               | 83.3  | 24.0       | 84.7  | 22.7       | 86.1  | 20.9       | 87.9  |
|                   |      |      | DB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 66.2   | 44.4  | 48.8  | 56.6  | 67.2  | 74.8  | 80.8  | 85.5  | 83.5  | 77.3  | 68.5  | 58.4  | 48.1 |
|                                                                                  | DBStd       | 15.80  | 8.47  | 10.37 | 9.53  | 8.61  | 6.29  | 5.31  | 4.11  | 4.83  | 6.48  | 6.82  | 8.15  | 6.69 |
|                                                                                  | HDD50       | 540    | 222   | 140   | 42    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 20    | 114  |
|                                                                                  | HDD65       | 2278   | 641   | 463   | 295   | 77    | 5     | 0     | 0     | 0     | 3     | 44    | 226   | 524  |
|                                                                                  | CDD50       | 6470   | 48    | 107   | 245   | 517   | 767   | 925   | 1101  | 1039  | 819   | 573   | 272   | 57   |
|                                                                                  | CDD65       | 2731   | 2     | 9     | 33    | 142   | 307   | 475   | 636   | 574   | 372   | 151   | 29    | 1    |
|                                                                                  | CDH74       | 28189  | 7     | 68    | 211   | 1089  | 2602  | 5009  | 8135  | 6598  | 3350  | 925   | 192   | 3    |
|                                                                                  | CDH80       | 12114  | 0     | 8     | 26    | 298   | 909   | 2096  | 4108  | 3028  | 1338  | 263   | 40    | 0    |
| Wind                                                                             | WSAvg       | 3.7    | 3.1   | 3.6   | 3.8   | 4.7   | 4.1   | 4.5   | 5.2   | 3.6   | 3.2   | 3.0   | 2.7   | 2.9  |
| Precipitation                                                                    | PrecAvg     | 61.6   | 2.7   | 3.3   | 6.3   | 7.1   | 8.5   | 9.6   | 6.3   | 6.7   | 3.5   | 2.6   | 3.0   | 2.0  |
|                                                                                  | PrecMax     | 83.7   | 7.8   | 6.7   | 14.3  | 12.2  | 15.6  | 13.9  | 13.8  | 12.5  | 6.7   | 10.0  | 8.5   | 6.1  |
|                                                                                  | PrecMin     | 42.1   | 0.4   | 0.6   | 3.1   | 2.4   | 4.2   | 5.3   | 1.1   | 2.2   | 1.1   | 0.0   | 0.0   | 0.2  |
|                                                                                  | PrecStd     | 9.8    | 1.6   | 1.7   | 2.3   | 2.4   | 2.4   | 2.4   | 3.2   | 2.9   | 1.4   | 2.2   | 1.9   | 1.6  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 74.2  | 80.4  | 83.2  | 90.0  | 93.2  | 95.7  | 99.4  | 99.7  | 96.5  | 90.1  | 85.0  | 73.1 |
|                                                                                  |             | MCWB   | 62.0  | 64.9  | 68.0  | 72.1  | 74.0  | 77.3  | 77.0  | 76.8  | 76.1  | 72.4  | 68.4  | 58.6 |
|                                                                                  | 2%          | DB     | 68.1  | 75.6  | 79.3  | 86.6  | 90.5  | 93.3  | 97.0  | 96.9  | 93.3  | 86.4  | 79.5  | 67.2 |
|                                                                                  |             | MCWB   | 56.7  | 62.6  | 65.6  | 71.2  | 73.7  | 77.1  | 77.1  | 76.7  | 75.8  | 70.3  | 65.1  | 56.2 |
|                                                                                  | 5%          | DB     | 63.1  | 71.2  | 76.0  | 83.5  | 88.0  | 91.3  | 95.1  | 94.4  | 90.8  | 82.7  | 74.8  | 62.9 |
|                                                                                  |             | MCWB   | 53.7  | 60.6  | 64.6  | 70.0  | 73.4  | 76.7  | 76.8  | 76.9  | 75.4  | 68.2  | 62.8  | 52.6 |
|                                                                                  | 10%         | DB     | 57.1  | 66.0  | 72.1  | 80.4  | 85.2  | 89.2  | 93.2  | 92.0  | 87.8  | 79.2  | 70.4  | 58.3 |
|                                                                                  |             | MCWB   | 49.8  | 57.7  | 62.3  | 68.9  | 73.0  | 76.4  | 76.8  | 77.0  | 74.8  | 67.3  | 60.7  | 51.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 63.0  | 66.6  | 70.3  | 74.6  | 78.6  | 80.1  | 80.9  | 80.9  | 79.2  | 75.7  | 70.4  | 63.9 |
|                                                                                  |             | MCDB   | 72.7  | 78.7  | 80.4  | 85.2  | 87.4  | 88.5  | 91.7  | 91.2  | 88.8  | 84.0  | 80.1  | 69.4 |
|                                                                                  | 2%          | WB     | 58.8  | 63.5  | 68.0  | 73.0  | 76.8  | 79.0  | 79.6  | 79.7  | 78.0  | 73.5  | 67.5  | 58.2 |
|                                                                                  |             | MCDB   | 66.3  | 73.8  | 76.3  | 83.2  | 85.2  | 88.2  | 90.1  | 89.5  | 87.6  | 81.5  | 76.0  | 63.4 |
|                                                                                  | 5%          | WB     | 54.5  | 61.2  | 65.7  | 71.8  | 75.6  | 78.1  | 78.9  | 78.8  | 76.9  | 71.7  | 65.2  | 54.6 |
|                                                                                  |             | MCDB   | 61.1  | 70.3  | 74.2  | 81.0  | 84.2  | 87.3  | 89.1  | 88.3  | 86.2  | 78.2  | 71.9  | 60.7 |
|                                                                                  | 10%         | WB     | 50.5  | 58.0  | 63.4  | 70.2  | 74.4  | 77.3  | 78.1  | 78.0  | 75.9  | 69.7  | 62.3  | 51.3 |
|                                                                                  |             | MCDB   | 56.7  | 65.2  | 70.9  | 78.7  | 82.7  | 86.4  | 88.2  | 87.2  | 84.7  | 75.9  | 68.9  | 56.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.5   | 10.5  | 11.4  | 12.4  | 12.4  | 11.2  | 12.3  | 12.2  | 12.2  | 12.3  | 12.5  | 11.3 |
|                                                                                  |             | MCDBR  | 21.8  | 20.9  | 19.6  | 17.8  | 17.2  | 13.7  | 14.9  | 16.1  | 16.9  | 19.1  | 20.8  | 21.7 |
|                                                                                  |             | MCWBR  | 12.1  | 10.8  | 9.2   | 7.1   | 5.9   | 4.2   | 3.6   | 3.7   | 4.6   | 6.6   | 8.8   | 11.3 |
|                                                                                  | 5% WB       | MCDBR  | 18.8  | 19.5  | 16.8  | 15.5  | 14.0  | 12.2  | 13.4  | 13.9  | 14.2  | 14.7  | 16.3  | 16.3 |
|                                                                                  |             | MCWBR  | 11.5  | 11.0  | 9.3   | 7.3   | 6.0   | 4.7   | 4.4   | 4.5   | 4.9   | 6.7   | 8.3   | 9.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.708  | 0.752 | 0.841 | 0.805 | 0.680 | 0.646 | 0.599 | 0.677 | 0.707 | 0.744 | 0.660 | 0.663 |      |
|                                                                                  | taud        | 1.476  | 1.437 | 1.338 | 1.427 | 1.674 | 1.754 | 1.869 | 1.672 | 1.597 | 1.486 | 1.597 | 1.562 |      |
|                                                                                  | Ebn at Noon | 174    | 181   | 175   | 188   | 214   | 220   | 230   | 212   | 200   | 180   | 185   | 177   |      |
|                                                                                  | Edh at Noon | 84     | 94    | 109   | 102   | 79    | 73    | 65    | 78    | 83    | 88    | 74    | 75    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 599    | 687   | 790   | 978   | 1163  | 1258  | 1616  | 1616  | 1434  | 1236  | 1059  | 840   | 723  |
|                                                                                  | RadStd      | 153    | 165   | 134   | 138   | 161   | 181   | 167   | 112   | 118   | 162   | 146   | 168   |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | -2.12  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (4 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page